

KAZ PP UPGRADE PROJECT

MANAGEMENT / STAFF EHS INDUCTION

General Instructions

- In case of Any Emergency during Induction, Follow the Exit Sign for Assembly Point.



During Induction:

- Keep Mobile Phone on SILENT Mode
- No Smoking



ON SILENT MODE PLEASE



KAZ PP Upgrade Project EHS Training

Content of EHS Induction

- Project Description
- Project EHS Policies
- Iraq Culture Information
- General Site Requirements
- Prohibited Items
- Hazards Associated With Working in live power stations
- Stop Work Authority
- Basic Security Requirements
- Emergency Response & Evacuation
- Road Safety
- Personal Protective Equipment's
- Heat Stress Management
- Fire & Use of Fire Extinguisher
- Safety in Camp / Office
- Waste Management System

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EHS Induction Objectives

At the end of this induction, you will:

- Be aware of Project EHS Policies
- Have full understanding of General Site Requirements
- Be aware of STOP Work Authority
- Be very familiar Hazards Associated With Working In live power stations
- Have full understanding of Road Safety Rules
- Be aware of Camp and Waste Management System
- Have full understanding of Emergency Preparedness and Response

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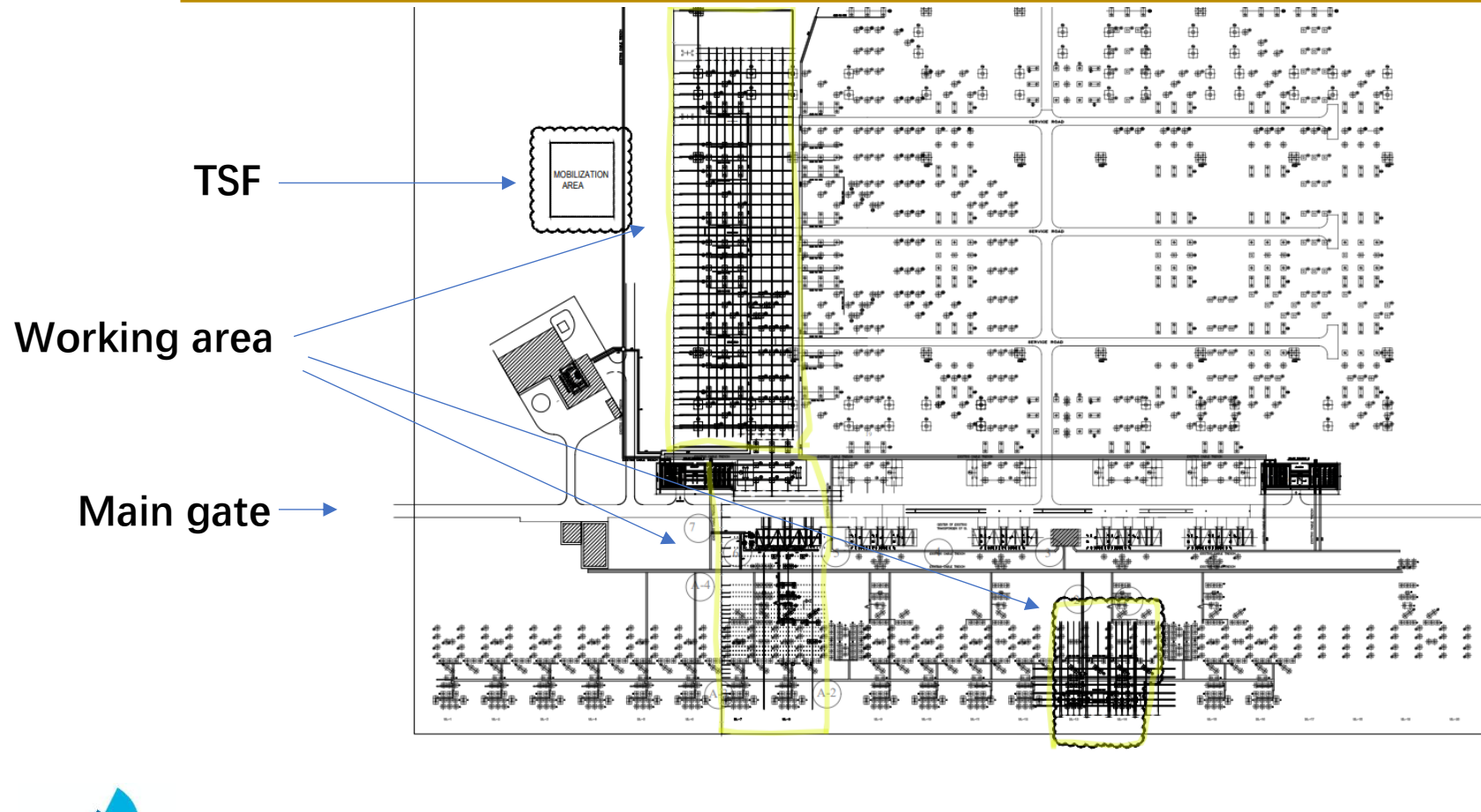
Project Description

The KAZ Power Plant Upgrade Project involves the upgrading and expansion of the electrical substation associated with the Khor Al Zubair Gas Turbine Power Plant in Basrah, Iraq.

The project includes civil works, installation, testing, commissioning and trial operation of a new 400/132/11 KV substation system, including one 250 MVA three-phase autotransformer bank, assembled from three single-phase transformer units, together with the associated 400 KV and 132 KV switchyards, protection and control systems, cabling, earthing, telecom, lighting and auxiliary systems

Project duration : 13.5 Month

KAZ Power Plant Upgrade Project - Overall Plot plan



Project HSE Policy



Health, Safety, and Environment (HSE) Policy & Strategic Objective

Policy Statement:

Our company is dedicated to maintaining the highest standards of Health, Safety, and Environmental (HSE) management throughout all oil and gas construction and services activities. We are committed to protecting our employees, contractors, and the communities in which we operate by minimizing risks, preventing accidents, and reducing environmental impacts. Compliance with all applicable regulations, industry standards, and best practices is fundamental to our approach.

We believe that all incidents are preventable, and we strive for continuous improvement in HSE performance by fostering a strong safety culture, ensuring workforce competency, and integrating sustainability into our operations.

Strategic HSE Objectives:

1. Health & Safety Objectives:

- **Zero Harm Goal:** Aim for zero fatalities, injuries, and occupational illnesses by enforcing strict safety controls.
- **Electrical Safety:** Implement stringent electrical safety measures to prevent electrocution, arc flashes, and other hazards.
- **Hazard & Risk Management:** Identify, assess, and mitigate all potential hazards associated with oil and gas construction activities.
- **Safety Leadership & Culture:** Promote a proactive safety culture where all employees and contractors take ownership of safety.
- **Training & Competency Development:** Ensure all personnel receive the necessary HSE training and certifications to perform their tasks safely.
- **Permit to work system:** Ensure strict permit to work procedures for live electrical work and high-risk construction task.
- **Emergency Preparedness & Response:** Establish and maintain robust site-specific emergency response plans for worksites Locations.
- **Incident Reporting & Learning:** Encourage transparent reporting of near misses and incidents to drive continuous improvement and prevent recurrence.



2. Compliance & Continuous Improvement Objectives:

- **Regulatory Compliance:** Ensure full adherence to industry-specific HSE regulations, guidelines, and best practices.
- **Performance Monitoring & Auditing:** Conduct regular HSE audits, risk assessments, and inspections to maintain compliance and identify areas for improvement.
- **Innovation & Technology:** Utilize new technologies and data-driven solutions to enhance HSE performance and risk management.
- **Stakeholder Engagement:** Collaborate with government agencies, industry partners, and local communities to strengthen HSE initiatives.
- **Continuous Improvement:** Regularly review HSE policies, update procedures, and implement lessons learned from past incidents.

Commitment & Accountability:

- **Leadership Commitment:** Management will demonstrate strong leadership by prioritizing HSE and providing necessary resources.
- **Employee & Contractor Responsibility:** Every individual is responsible for maintaining a safe work environment and adhering to HSE policies.
- **Continuous Improvement:** We will regularly review and enhance our HSE policies, objectives, and performance through audits and feedback.

By embedding HSE excellence into our core values, we aim to achieve operational efficiency, safeguard our workforce, and protect the environment while delivering world-class high and medium voltage substations construction projects.

AL-MIAL CEO

Name:

Hussein M.frah



AMC HSE POLICY AIMED:

1. ZERO INJURY.

2. ZERO POLLUTION

3. ZERO ACCIDENT



**ZERO
ACCIDENT**



**ZERO
INJURY**



**ZERO
POLLUTION**

Iraq Culture Information

Basic greetings in Arabic.

“ *Asalaamu alaikum*” (peace be with you)

“ *Wa alaikum salaam*” (and peace be unto you)

“ *Merhaba*” - Hello

“ *Shukran*” – Thank You

- **Make eye contact:** (But not with women), smile and say “*Merhaba*”
- **Good friends of the same sex:** May greet each other with a handshake and a kiss on the cheek.
(*not considered ill mannered as Iraqis are highly demonstrative people*)
- **Shaking hands with ladies:** Men should wait to see if the woman extends her hand (very rare).
- **Introductions:** Expect to be introduced to each person individually at all meetings
- Always use **Right hand**:
 - Eating
 - Greeting
- **DO NOT** Enter Mosque or prayer room without removing shoes.
- **DO NOT** Eat, drink or smoke in public place during month of Ramadan.
- **DO NOT** Remove/take down any Religious flags/Symbols
- **DO NOT** “Hush SSSSHHHH” anyone.

Note: Offering to shake with the left hand may be perceived as an insult!!



General Site Requirements

- EHS Induction before starting any work.
- PTW System to be implemented.
- EHS supervision must be at the work place when work is being conducted **(No HSE = No work)**
- **Wear Mandatory Personal Protective Equipment (PPEs)** Coveralls, safety shoes, hard hats, safety goggles, Hand Gloves, High Visibility Vest.



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- **EHS Trainings.**
- **Follow the STOP WORK AUTHORITY**, if any unsafe act or unsafe conditions.
- **Aware of Emergency Response Procedure.**
- **All drivers and operators must have valid Iraq license.**
- **All Vehicles must be inspected and 3rd party certified.**

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- Only Reverse Parking in designated parking areas .
- Follow Speed Limit (5 KM\H) .
- All equipment and heavy vehicle movement controlled by flagman.
- Stop the work in case of emergency.
- Report all Near Misses/accidents.



Electrical Safety Rules

1. Treat every electrical circuit as energized until proven otherwise.
2. Isolate the power source before starting any work.
3. Apply Lock Out/Tag Out (LOTO) to prevent accidental re-energization.
4. Verify the absence of voltage using an approved testing device.
5. Wear the required electrical PPE (arc-rated clothing, insulated gloves, face shield, safety helmet, and safety footwear).
6. Use insulated tools and equipment that are suitable for the voltage level.
7. Maintain safe approach distances from live electrical equipment.

ELECTRICAL SAFETY RULES

1 Treat every electrical circuit as energized until proven otherwise. 	2 Isolate the power source before starting any work. 	3 Apply Lock Out/Tag Out (LOTO) to prevent accidental re-energization. 	4 Verify the absence of voltage using an approved testing device. 	5 Wear the required electrical PPE.
6 Use insulated tools and equipment that are suitable for the voltage level. 	7 Maintain safe approach distances from live electrical equipment. 	8 Never work on live electrical equipment unless it is absolutely necessary and properly authorized. 	9 Keep electrical panels and access routes clear and free from obstructions. 	10 Report damaged electrical equipment, exposed wires, or unsafe conditions immediately.

GOLDEN RULE
ISOLATE – LOCK OUT – TAG OUT – TEST BEFORE TOUCH

ELECTRICAL HAZARD

ARC FLASH HAZARD

FIRE HAZARD

TOXIC FUMES HAZARD

HOT SURFACE HAZARD

TRIP HAZARD

EARTH CONNECTION

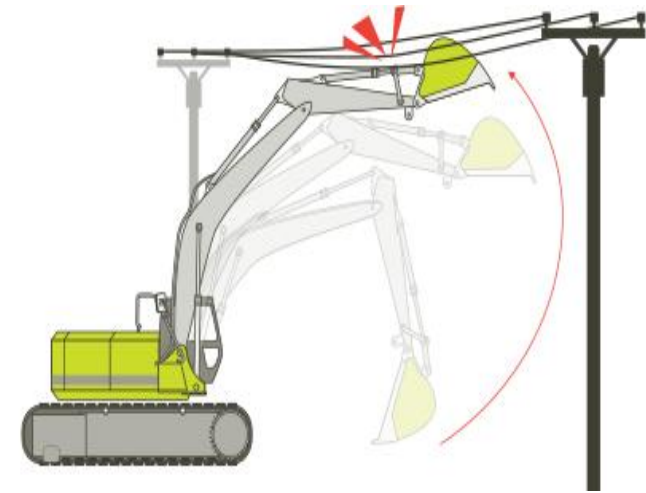
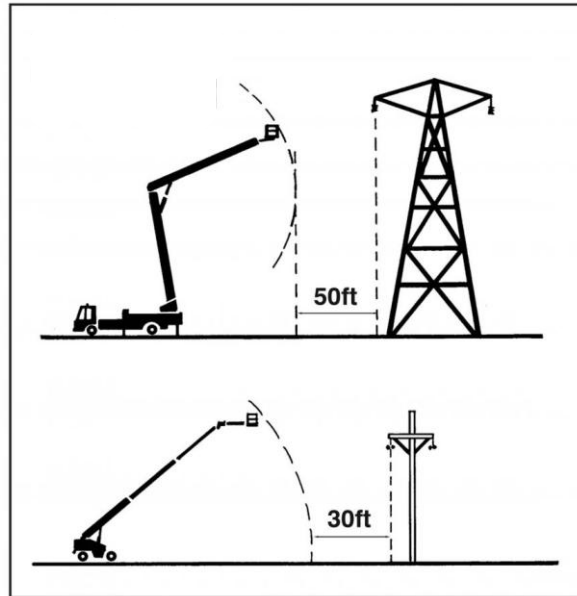
8. **Never work on live electrical equipment** unless it is absolutely necessary and properly authorized.
9. **Keep electrical panels and access routes clear** and free from obstructions.
10. **Report damaged electrical equipment, exposed wires, or unsafe conditions immediately.**



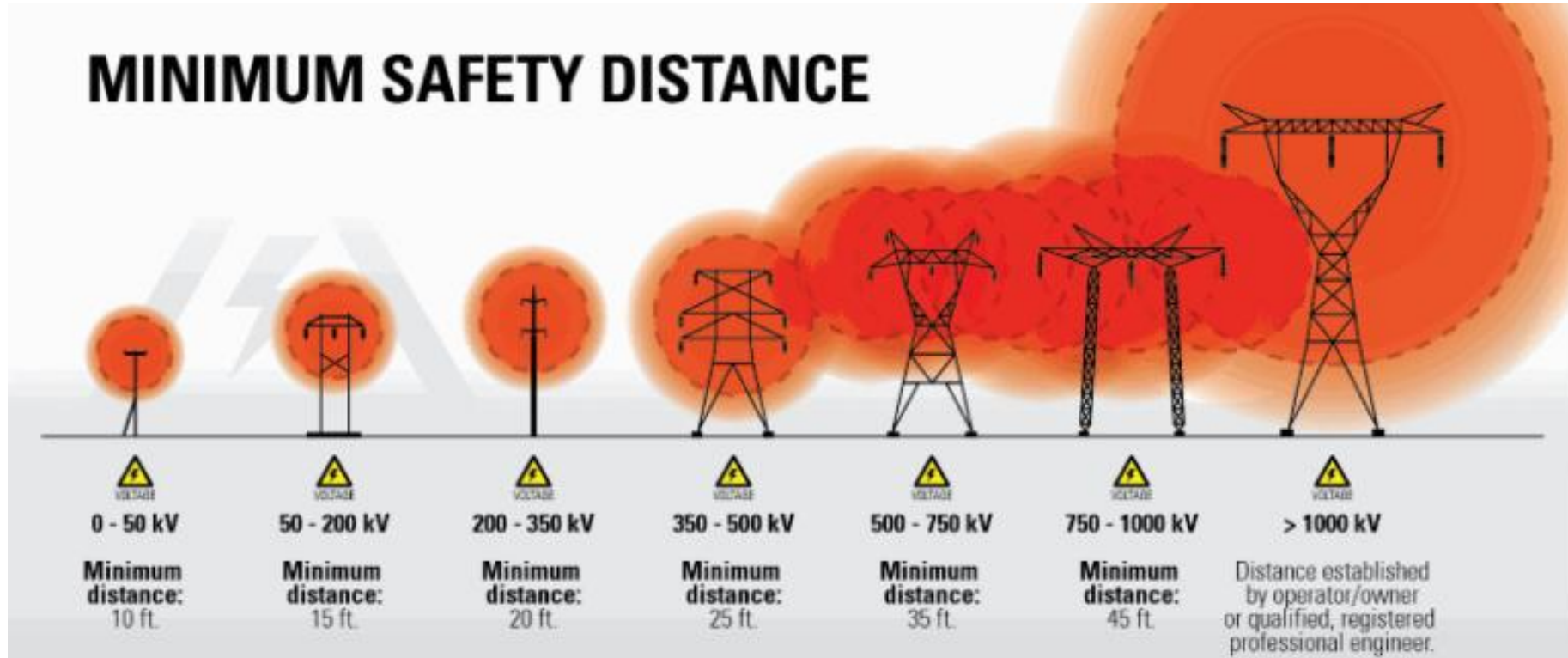
Golden Rule

"Isolate – Lock Out – Tag Out – Test Before Touch."

Electrical Safe Distance



Check what above and near you



Prohibited Items



- Alcohol



- Narcotics (Drugs)



- Fire arms and ammunition



- Smoking (Only in Designated area)



- Knives



- Camera/Photography
in Restricted Area



- No Mobile Phones in
Restricted Area

Hazards – Associated with Working in live power stations



Electrical Fire



Electrical Arc





Electrical Shock



- Poisonous Reptiles/Insects Hazards.



Poisonous Creatures

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GOLDEN MOTTO

Always
Remember

DO IT SAFELY, OR DON'T DO IT.

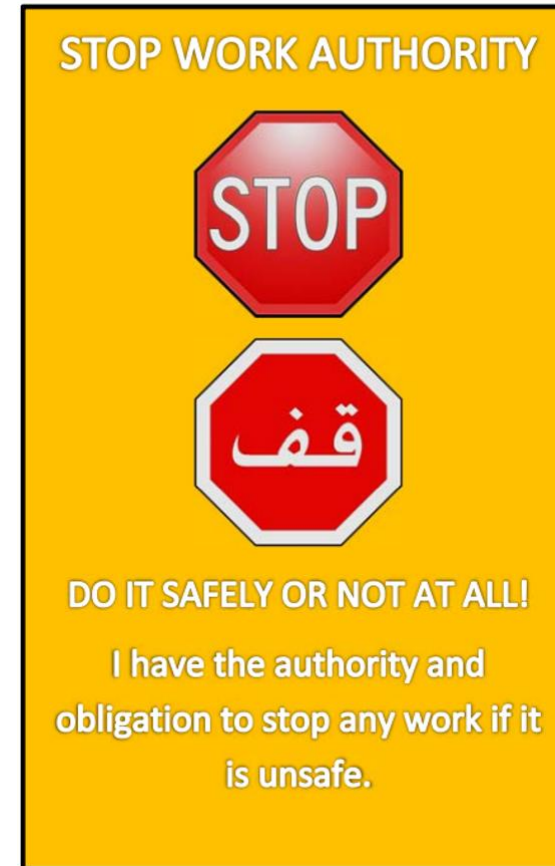


AMC EHS MOTTO

Stop Work Authority

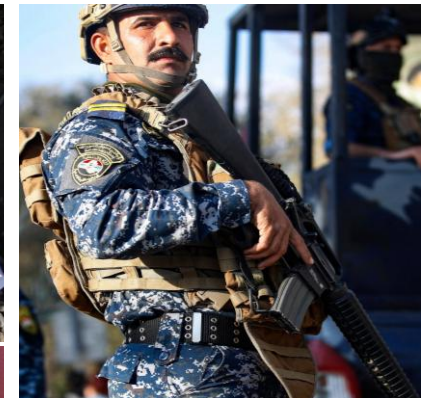
Protocols Instructions – Steps

- Stop the activity if appropriate when imminent danger or risk exists
- Intervene directly with the personnel involved
- Escalate the concern to the supervisor or responsible
- Positive feedback shall be given to affected employees
- Report formally and documented



Project Security and Police Instructions

- 1- Camps & Site are protected Iraqi Police.
- 2- Movement only with prior authorization.
- 3- Compulsory use Seat belts – Driving Rules
- 4- Passport with valid Visa
- 5- ID Card for local People
- 6- Get Pass shall be with you always (**security clearance**)
- 7- Don't be afraid of the police when they carry some weapons
- 8- Always be cooperative with the security authorities. And remember that they are here to protect us.



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Emergency Response Evacuation

Emergency is defined as a serious, unexpected and often dangerous situation requiring immediate action.

Classification of Emergencies:-

- Level -1 Emergency
- Level -2 Emergency
- Level -3 emergency

Level 1
Emergency

Level 2
Emergency

Level 3
Emergency

Level -1 Emergency Notification

- An Event/ Incident can be managed at the site level with personnel / equipment's on-site resources
- Key words: Minor health effect/injury, Minor damage, Minor Impact



**Level 1
Emergency**

Level -2 Emergency Notification

- An Event/ Incident that can be managed locally with support from COMPANY or external local authorities
- Key words: Major health effect/injury, Fatality, Local damage, Local Impact

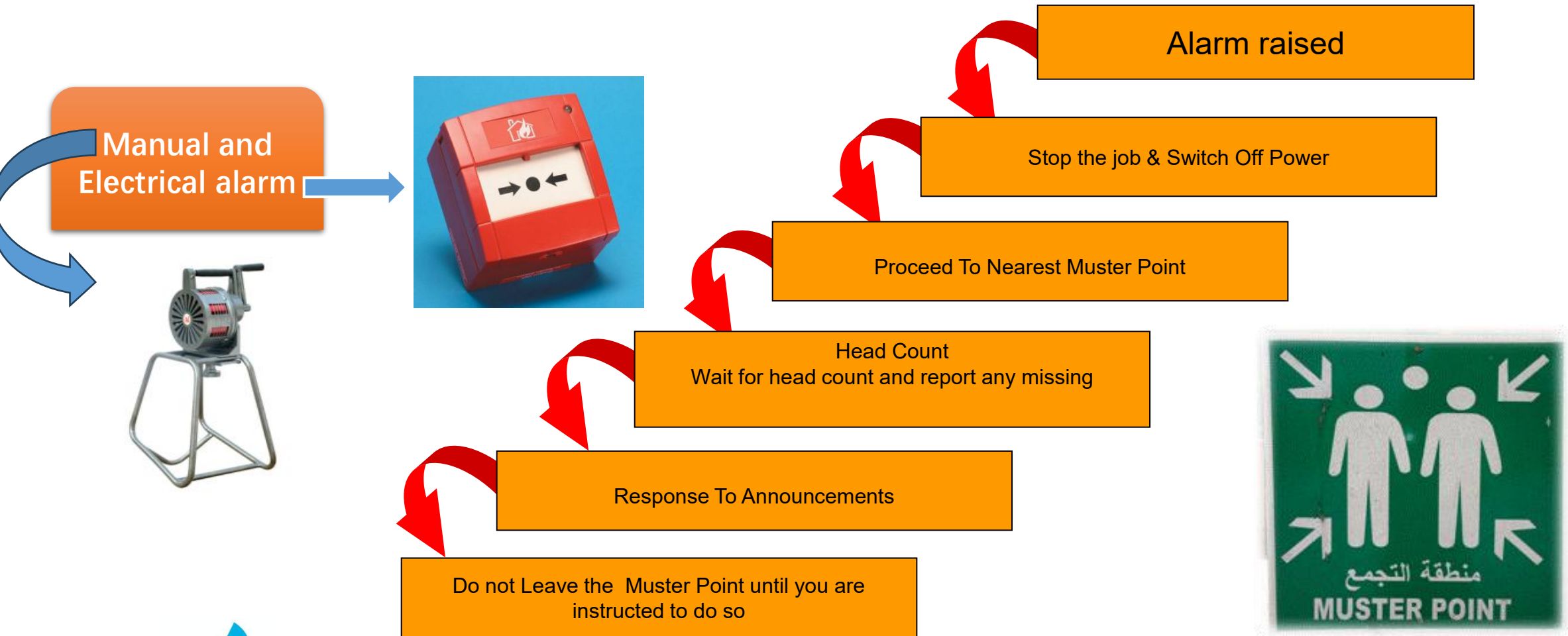
Level 2
Emergency

Level -3 Emergency Notification



- A major event /incident which is beyond the contractor control and requires resources beyond its scope, needs involvement from the COMPANY due to the border implication.
- Level 3 can be also a crisis situation; it depends on an event or sequence of extreme concern or danger and is able to disrupt contractor capability to bring the situation under control.
- Key Words : Multiplies fatalities, Extensive damage and Major international impact

Emergency Response & Evacuation



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Emergency Reporting Procedure

In case of any emergency undertake below mentioned actions.

- Call AMC Emergency Contact Number **0785-285-7404 & 0786-162-9706**
- Haider Katea : 0770-731-3604 (**First Ader**)
- Don't hesitate to call the emergency number **911** if there's any serious threat, even in your home or anywhere else.
- Tell your name and company name
- Tell your exact work location
- Talk about incident



Driving Safety

Road Safety General Instructions

- Driving restricted during Adverse Weather (FOG, Rain & Dusty)
- Pre-Use Checks shall be done by daily basis
- Wear seat belts
- Follow project SPEED LIMIT 5 K/H
- Driver shall never use MOBILE PHONE
- Drivers must have driving license



Personal Protective Equipment

- Hard Hats



- Safety Glasses/ Goggles



- Safety gloves



- Safety shoes



- Safety coverall



- High Visibility Vest



- Face Mask (Not Mandatory)



Heat Stress Management

HEAT INDEX (TEMP)	HEAT CATEGORY	FLAG COLOR	WORK-REST CYCLE		AMOUNT OF WATER INTAKE
			LIGHT WORK	HEAVY WORK	
46°C Above	Very High to Extreme	RED	45 Minutes Work & 15 Minutes Rest / Hour	20 Minutes Work & 40 Minutes Rest / Hours	1.2 Liter or Above / Hour
41°C to 46°C	High	Orange	Light Work shall be Continued	45 Minutes Work & 15 Minutes Rest / Hour	1.2 Liter or Above / Hour

41°C to 46 °C

HIGH RISK

- Working alone to be avoided
- Light Work- Drink water 1 – 1.2 liters/hr.
- Heavy works- Work/rest cycle 45 min Work – 15 min rest

46 °C Above

VERY HIGH RISK

- No Person to work alone
- Strict work/rest cycling required
- Light Work:- 45 min work – 15 min rest
- Heavy Work:- 20 min work – 40 min rest

Fire & Use of Fire Extinguishers

Fire & Use of Fire Extinguishers

☑ THERE ARE FOUR CLASSIFICATIONS OF FIRE & FIRE EXTINGUISHER:

- Class – A : Wood, paper, cloth, rubber, some plastics
- Class – B : Gasoline, kerosene, paint, propane.
- Class – C : Fire Electrical equipment.
- Class – D : Metals like magnesium, titanium, sodium
- Witte Chemical fire Extinguisher
- Class – A : DCP
- Class – B : DCP & Foam.
- Class – C : CO2 & Power Switch Off.
- Class – D : DCP



Fire & Use of Fire Extinguishers

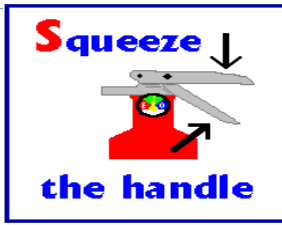
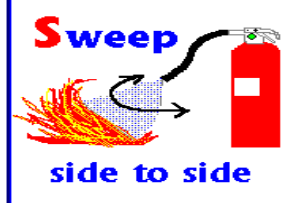
Fire Fighting

PASS method of fighting fires

- ☐ Pull the pin
 - ☐ Aim at the base of the fire
 - ☐ Squeeze the handle
 - ☐ Sweep at the base of the fire
- Hold the extinguisher upright

Extinguishers should be available on visible location

Maintain in good working order
Do not put yourself in danger

P	<u>PULL</u> the pin, this unlocks the lever and allows you to discharge the extinguisher	
A	<u>AIM</u> low: point the extinguisher nozzle (or hose) at the base of the fire	
S	<u>SQUEEZE</u> the lever above the handle: this discharges the extinguishing agent	
S	<u>SWEEP</u> from side to side moving carefully toward the fire	

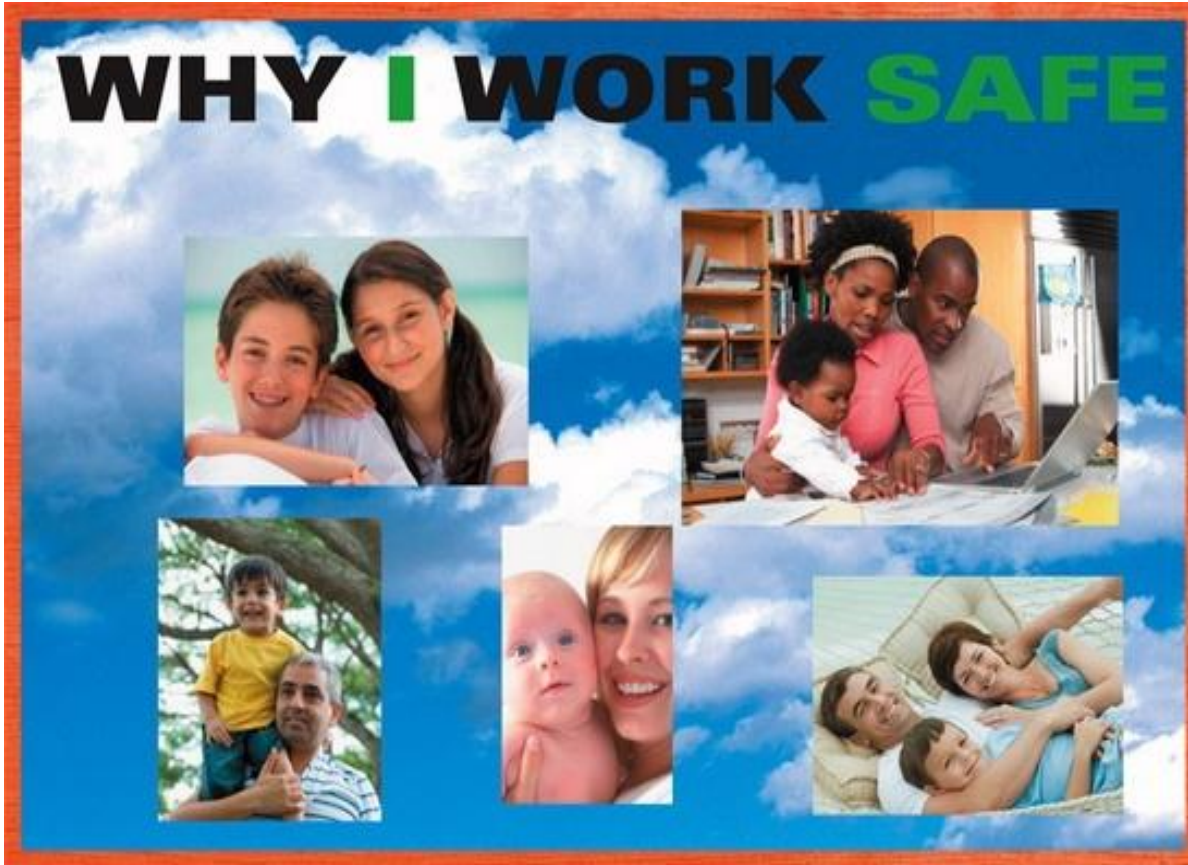
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Near miss / Incident Reporting

1. Any unsafe condition / action & Near miss shall be report to the **EHS Department**.
2. Investigation team will be set up for significant incidents by Contractor,
COMPANY and Sup contractor
3. Investigation team will issue Investigation Report containing all root causes
4. Learning from incidents (LFI) will be issued as lesson Learned (LL).

**DO NOT FORGET TO GET
YOUR INDUCTION CARD**





Thanks

*To ensure that everybody returns home
without injury & Safely*

